

Hybrid Autonomous Navigation in BEV with Route Optimization for Crop Monitoring

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Abstract

This literature review supports the Master’s thesis *Hybrid Autonomous Navigation in BEV with Route Optimization for Crop Monitoring* (original Spanish title: *Navegación Autónoma Híbrida en BEV con Optimización de Rutas para Seguimiento de Cultivos*), which proposes a modular autonomous agricultural vehicle system combining multimodal Bird’s-Eye View perception, hybrid classical-plus-learned path planning, and crop-health computer vision. The review synthesises research across three thematic domains: (1) transformer-based BEV perception architectures including BEVFormer, BEV-Fusion, BEVDet, and PETR, together with 3D occupancy prediction and their academically cautious transfer to agricultural settings; (2) hybrid path planning combining Hybrid A*, lattice planners, imitation-learned trajectory scoring, safety validation layers, and Model Predictive Control, with classical coverage planning as the operational backbone; and (3) agricultural computer vision addressing crop disease detection, weed segmentation, domain-adaptation challenges, and benchmark datasets including PlantVillage, DeepWeeds, and Agriculture-Vision. A system-level integration section proposes a staged, prototype-appropriate modular architecture. Recommended experimental methodology accounts for the absence of field BEV maps by prescribing simulation-first validation, domain-adaptation protocols, and incremental deployment on available hardware. Reinforcement learning is identified explicitly as a direction for future research rather than a component of the proposed system. Research gaps, thesis contributions, and evaluation protocols suitable for an academic prototype are articulated throughout.

1 Scope and Method

1.1 Research Scope

This review covers literature directly relevant to an autonomous agricultural vehicle capable of (a) perceiving its environment as a unified Bird’s-Eye View (BEV) representation from multiple sensors, (b) planning and executing coverage routes that respect vehicle geometry and operational constraints, and (c) identifying crop-health patterns in real time. The thesis scope is deliberately bounded to a Minimum Viable Prototype (MVP) architecture validated first in simulation, then on a physical platform comprising a Raspberry Pi 3B, dual cameras, IMU, wheel encoders, and GNSS [1]. Production-grade real-time BEV inference, multi-vehicle fleet management, and autonomous implement docking are treated as directions for future development [2].

1.2 Methodology

Sources were selected to represent foundational and current-state-of-the-art work in each thematic domain. Priority was given to peer-reviewed conference and journal publications indexed on arXiv and major digital libraries, as well as official dataset repositories. Claims are synthesised thematically rather than source-by-source, and areas of agreement or disagreement across sources are noted explicitly. The

review explicitly distinguishes established results from single-source claims and from thesis-level projections.

2 Multimodal BEV Transformer Perception

2.1 The BEV Representation Paradigm

The Bird’s-Eye View representation has become the dominant spatial encoding in autonomous driving perception because it provides a metric top-down map in which vehicle geometry, obstacle extents, and traversability are naturally expressed. Rather than reasoning in perspective-view image space, a BEV encoder converts heterogeneous sensor observations—camera images, LiDAR point clouds, and radar returns—into a shared planar or volumetric feature grid aligned with world coordinates. This alignment is particularly valuable for path planning: the planning module can directly query the BEV grid for free space and obstacle locations without further geometric transformation [2, 3].

The fundamental challenge is *view transformation*: projecting 2D image features into a 3D or BEV coordinate frame requires either explicit depth estimation, geometric lifting, or implicit attention-based querying. Different architectures address this challenge through distinct mechanisms, each with well-documented trade-offs in accuracy, computational cost, and robustness to sensor miscalibration [4-6].

2.2 Foundational Camera-Only Architectures

2.2.1 BEVFormer

BEVFormer [5, 6] introduced a spatiotemporal transformer architecture in which a fixed grid of learnable BEV queries interacts with multi-view image features through two purpose-built attention mechanisms. The *spatial cross-attention* module associates each BEV query with regions of interest across all camera views, implementing a form of implicit epipolar geometry without requiring explicit depth prediction. The *temporal self-attention* module recurrently fuses BEV feature maps from previous time steps, operating analogously to an RNN hidden state to propagate motion and occlusion cues across frames [6]. This architecture was reported to achieve 56.9% NDS on the nuScenes test set, substantially outperforming prior methods such as DETR3D, and is particularly noted for improved velocity estimation and object recall under low-visibility conditions [5]. Subsequent variants have extended BEVFormer with gated transformer layers for instance prediction (BEVPredFormer [7]) and with learning-first view attention to relax sensor deployment constraints (ViewFormer [8]).

2.2.2 BEVDet and BEVDet4D

BEVDet and its temporal extension BEVDet4D employ the Lift-Splat-Shoot (LSS) module for 2D-to-3D feature transformation, offering higher computational efficiency than the attention-heavy BEVFormer design [9, 10]. BEVDet4D augments the

single-frame approach by preserving and concatenating BEV feature maps from preceding frames, capturing temporal clues for velocity estimation and handling of briefly occluded objects [9–12]. The LSS mechanism explicitly discretises the camera frustum and distributes pixel-level features into a voxel grid using predicted categorical depth distributions, which makes the depth prediction quality a critical bottleneck [9, 10].

2.2.3 PETR and StreamPETR

PETR (Position Embedding Transformation) replaces the complex online 2D-to-3D feature sampling of earlier methods with a coordinate-based approach [13, 14]. The camera frustum space is discretised into a 3D mesh-grid, transformed into world coordinates using calibrated camera parameters, and these coordinates are encoded into the 2D image features extracted by a backbone network. The resulting 3D position-aware features allow object queries to perform end-to-end detection directly in the 3D domain through a transformer decoder, avoiding the inaccurate reference-point sampling that limited prior detectors [14]. StreamPETR extends PETR with object-centric temporal modelling, enabling robust multi-frame reasoning and forming part of ensemble solutions for occupancy prediction benchmarks [10, 15–17].

A practically important line of research concerns the deployment of PETR on resource-constrained hardware. Quantisation to INT8 precision—relevant for embedded platforms such as the STM32 or edge inference accelerators—causes “quantisation collapse” due to extreme magnitude disparities between image features and the camera-ray position embeddings [18, 19]. Proposed mitigations include anchor-based position embedding re-parameterisation and cascaded linear lookup tables for non-linear function approximation [18, 19], which are relevant considerations for any agricultural prototype targeting embedded deployment.

2.3 Multi-Sensor Fusion: BEVFusion

BEVFusion [4, 12] addresses the fundamental limitation of point-level fusion approaches, which augment LiDAR point clouds with camera features. Such approaches are semantically lossy: for a 32-beam LiDAR, camera-to-LiDAR projection discards a substantial fraction of camera features, hindering performance on semantic tasks such as drivable-region segmentation [4, 20, 21]. BEVFusion instead maps both modalities into a shared BEV feature space, preserving the geometric structure of LiDAR and the semantic density of camera imagery simultaneously [4, 20]. A critical engineering contribution is the optimised BEV pooling kernel, which achieves over 40× speedup in view transformation compared to prior implementations, enabling near-real-time multi-modal fusion [4, 12, 22].

Subsequent research has extended BEVFusion to handle sensor failure scenarios. “Single-Branch” (SB-BEVfusion) variants disentangle the camera and LiDAR processing streams, ensuring that the camera branch remains functional during LiDAR malfunctions and vice versa—a critical safety property for any operational deployment [23]. Camera-radar fusion variants employing Radar Gaussian Expansion

further demonstrate that radar measurements, distributed as Gaussian distributions within BEV voxels, can reduce association errors and improve detection of dynamic objects under adverse weather conditions [11].

2.4 Temporal Fusion and Spatiotemporal Modelling

Temporal modelling is identified consistently across the literature as essential for accurate velocity estimation, prediction of briefly occluded objects, and overall scene understanding [11, 17]. The key design choice is the granularity at which temporal information is aggregated: BEVFormer operates at the BEV feature level [5, 6], BEVDet4D at the concatenated BEV-frame level [11, 12, 16], and StreamPETR at the object-query level [10, 15–17]. Each level entails different computational costs and inductive biases. For agricultural vehicles operating at low speeds with relatively static environments, single-frame BEV representations may suffice as a starting point, with temporal fusion added as a second-stage improvement. RoPETR integrates Multimodal Rotary Position Embedding into the StreamPETR framework, decomposing temporal, height, and width components for improved motion consistency across frames [24].

2.5 BEV Occupancy and Free-Space Prediction

Beyond object detection, BEV occupancy prediction provides a voxel-level semantic description of whether each spatial cell is occupied, free, or unobserved [25, 26]. This representation is more robust to irregular or non-standard obstacles—a particularly relevant property in agricultural settings where objects such as fallen branches, irrigation pipes, or animals do not conform to the object classes of road-vehicle datasets.

The Occ3D-nuScenes benchmark [27–29] provides a standardised evaluation framework with a 200 by 200 by 16 voxel grid at 0.4 m resolution, covering $[\pm 40]$ m in X/Y and $[-1, 5.4]$ m in Z, annotated with 16 semantic classes plus a “General Objects” catch-all category [28]. The benchmark employs an automated label generation pipeline comprising voxel densification via multi-frame aggregation, occlusion reasoning through ray-casting, and image-guided voxel refinement to eliminate 3D–2D misalignment artefacts [28].

Multiple architectural strategies have been proposed to balance accuracy and inference efficiency. TPVFormer decomposes 3D space into three orthogonal planes (BEV, front, and side views), reducing the computational cost of lifting camera features to 3D [25, 30]. SparseOcc adopts a fully sparse paradigm using mask-transformer queries to avoid dense 3D convolutions, achieving real-time inference at 17.3 FPS and introducing the **RayIoU** metric that evaluates occupancy by casting rays into the predicted volume—addressing ambiguities of traditional voxel-level mIoU in areas not covered by LiDAR sweeps [31]. GSD-Occ employs a hybrid BEV-voxel representation, separating semantic and geometric branches to decouple semantic classification from geometric refinement [26]. For deployment without precise calibration, REO employs vanilla cross-attention to implicitly model spatial correspondence, demonstrating robustness to calibration noise [32]—a property highly desirable in field robotics

where calibration may drift due to vibration and mud accumulation.

The OccTransformer achieves competitive results by combining stronger image backbones with a 3D U-Net head that preserves fine-grained spatial information, and by ensembling results from BEVDet and SurroundOcc together with StreamPETR for dynamic object classes [10, 15].

2.6 Geometric Projection and Coordinate Calibration

All BEV perception methods depend critically on accurate intrinsic and extrinsic calibration between sensors. For camera-LiDAR-radar-IMU systems, the calibration pipeline must account for mechanical mounting offsets, time-stamp synchronisation, and dynamic distortions introduced by vehicle vibration [3]. In the autonomous driving domain, calibration is typically performed offline, but agricultural vehicles operating in dusty and vibration-prone environments require periodic recalibration or self-calibrating mechanisms. The REO framework [32] demonstrates that learning-based, calibration-agnostic BEV transformation is feasible, suggesting a path toward agricultural robustness.

2.7 Transfer of BEV Perception to Agricultural Robotics: A Cautious Appraisal

The overwhelming majority of BEV perception research is evaluated on urban road-vehicle datasets such as nuScenes [27, 29], which feature structured road environments, standardised object categories (pedestrians, vehicles, cyclists), and high-density sensor rigs unavailable in agricultural prototypes. Several structural differences limit direct transfer:

- **Object class mismatch:** Agricultural obstacles include trees, posts, animals, and implements, none of which appear as training categories in nuScenes [3].
- **Sensor configuration:** Road benchmarks typically assume six or more cameras with known baselines and calibrated LiDAR rigs, whereas the proposed thesis prototype uses two cameras and potentially a single frontal LiDAR [1, 3].
- **Environmental texture:** Agricultural scenes feature repetitive texture (crop rows), lack lane markings, and exhibit strong seasonal appearance changes [33, 34].
- **GNSS availability:** Under dense crop canopies, GNSS signals attenuate significantly, requiring heavier reliance on visual or LiDAR-based localisation [34, 35].

A constructive transfer strategy begins with pre-training BEV backbone features on large road-vehicle datasets, using domain randomisation and domain adaptation techniques to bridge the appearance gap, and then fine-tuning on synthetic or

real agricultural data. The agricultural robotics literature confirms that GNSS-RTK provides centimetre-level accuracy in open fields [33] but degrades under canopy, necessitating sensor fusion with IMU, wheel odometry, and optionally visual SLAM [36–38]. For a thesis-level prototype, a simplified occupancy representation derived from camera-based depth estimation fused with LiDAR geometry is a realistic starting point, deferring full multi-modal BEV transformer deployment to a more instrumented platform.

Table 1: Comparison of BEV perception architectures relevant to the thesis. Agricultural suitability is assessed with reference to the target embedded platform and environment.

Method	Input Modalities	Key Innovation	Temporal Fusion	Agricultural Suitability
BEVFormer [5, 6]	Multi-camera	Spatiotemporal BEV queries; spatial cross-attention	Recurrent temporal self-attention	Moderate; high compute; needs calibration
BEVDet4D [9, 10]	Multi-camera	LSS lifting; efficient pipeline	Frame concatenation	Moderate; lighter than BEVFormer
PETR / StreamPETR [13, 14]	Multi-camera	3D position-aware features; no explicit lifting	Object-centric temporal queries	Moderate; quantisation challenges for embedded [18, 19]
BEVFusion [4, 12]	Camera + LiDAR	Shared BEV fusion; 40× pool speedup	Inherited from per-modality encoders	High when LiDAR available; sensor-failure robustness via SB variant [23]
TPVFormer [25, 30]	Multi-camera	Tri-perspective decomposition	Implicit temporal	Good for low-compute platforms
SparseOcc [31]	Multi-camera	Fully sparse; RayIoU metric	Sparse temporal queries	Good real-time; novel metric avoids label ambiguity

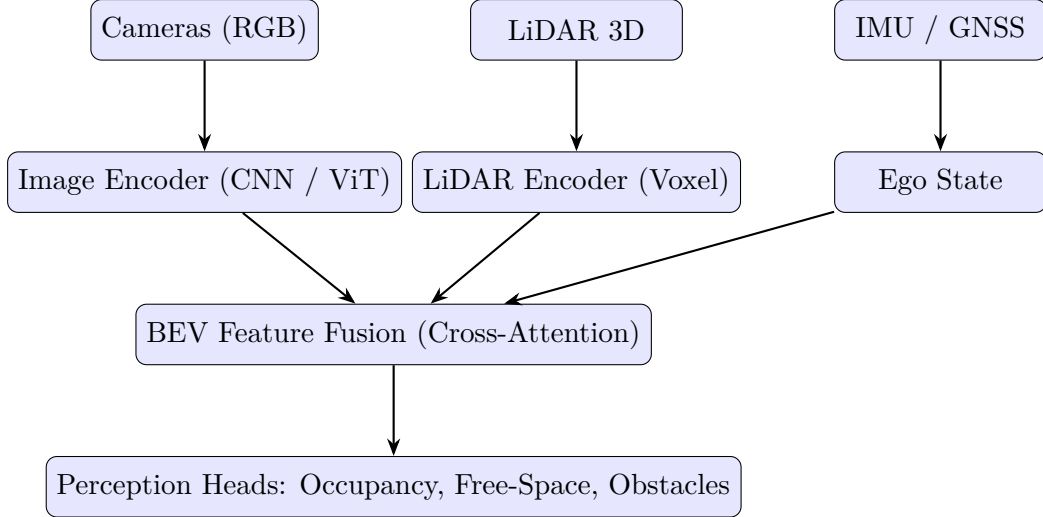


Figure 1: Generalised BEV perception pipeline applicable to the thesis architecture. Camera and LiDAR features are lifted to a shared BEV feature space via attention-based fusion, producing occupancy, free-space, and obstacle representations for the downstream planner.

3 Hybrid Path Planning for Agricultural Vehicles

3.1 The Case for Hybrid Rather Than Pure Learning-Based Planning

A central design choice of this thesis is the use of a *hybrid* planning architecture in which a classical deterministic coverage planner provides the global trajectory skeleton, an optionally learned local planner refines short-horizon decisions, and an MPC layer executes the trajectory while enforcing kinematic and safety constraints [2, 3]. This choice is motivated by a consistent finding in the autonomous driving planning literature: learning-based planners excel at generating human-like trajectories in distribution but exhibit unpredictable and difficult-to-bound failures in out-of-distribution scenarios [39–41]. In safety-critical agricultural machinery, an error is not merely a “visual artefact”: it can damage crops, injure animals, or cause a serious mechanical accident [3]. Classical planners with formal safety guarantees remain the appropriate backbone, with learned components providing adaptability within a verified safe envelope.

The hybrid planning paradigm—integrating learned trajectory proposals or scoring with optimisation-based or search-based safety verification, followed by MPC execution—is explicitly identified as a current research trend in trajectory planning surveys covering 2020–2025 [42–44]. The thesis adopts this trend in the specific context of agricultural coverage, where additional constraints including minimum turning radius, implement width, soil compaction, and crop-row alignment are imposed on the planner [45–47].

3.2 Classical Coverage Path Planning

3.2.1 Boustrophedon and Cellular Decomposition

Boustrophedon cellular decomposition partitions a field into convex sub-cells, each of which is covered by a set of parallel swaths aligned with the implement working width [47]. This approach provides a deterministic guarantee of complete coverage when the vehicle tracks each swath within a bounded lateral error, and its simplicity makes it the natural baseline for the thesis system [1]. The primary operational parameters are swath width (dictated by the cutter or sprayer implement), overlap fraction (to compensate for wheel-track variation and achieve the target 98% coverage with $\leq 2\%$ redundant overlap), and headland width (required for turn manoeuvres) [1, 45].

Coverage Path Planning (CPP) is formally defined as computing the path for a robot that guarantees coverage of an entire target area [48, 49]. In agricultural robotics, CPP must additionally optimise for non-working travel distance (headland turns and inter-swath transitions), which can represent a significant fraction of total operating time in fields with irregular boundaries or internal obstacles [47]. Multi-robot CPP approaches partition fields into convex sub-regions distributed among multiple agents to reduce mission time [48], though single-vehicle coverage is the thesis focus.

3.2.2 Routing Optimisation Beyond Pure Boustrophedon

For fields with multiple zones requiring treatment (e.g., weed cluster locations identified by the crop-health perception module), scheduling the inter-zone visit sequence as a Travelling Salesman Problem (TSP) instance and applying localised coverage within each zone significantly reduces energy consumption compared to naive full-field coverage [50]. Boundary-shifting algorithms ensure complete coverage within each sub-region while the TSP model orders inter-region transitions efficiently [50]. Intelligent optimisation algorithms including Genetic Algorithms (GA), Particle Swarm Optimisation (PSO), and Ant Colony Optimisation (ACO) have been applied to CPP with irregular field boundaries and internal obstacles [51], and can be incorporated as the global planner without introducing the unpredictability of learning-based approaches.

3.3 Hybrid A*, Lattice Planners, and RRT Variants

Graph-search and sampling-based planners generate kinematically feasible paths by explicitly modelling vehicle non-holonomic constraints. Hybrid A* extends the standard A* search to continuous state spaces by considering vehicle kinematics at each expansion step, making it suitable for generating headland turn manoeuvres and paths in cluttered environments [43, 47, 52]. Recent agricultural research has adapted Hybrid A* as a *path-tracking* planner functioning as a path smoother rather than a pure global planner, incorporating whole-body collision avoidance and specialised cost functions that minimise lateral deviation from crop-row reference lines to prevent crop damage [45, 46].

Lattice planners discretise the vehicle’s motion space into a set of pre-computed motion primitives (arcs, straight segments, and turn-in-place manoeuvres), providing a structured search space that enforces kinematic feasibility by construction [43]. Rapidly-exploring Random Tree (RRT) and its variants are particularly suited to high-dimensional configuration spaces encountered in orchard navigation or when tracking a tractor-trailer system with articulated dynamics [47].

Quadratic programming (QP) over safe corridors provides an alternative: convex regions of free space are computed around an initial rough path, and a high-order continuous trajectory is optimised within these corridors, yielding smooth paths with bounded curvature and jerk that are easier for the MPC controller to track accurately [53]. The Frenet frame transformation decouples longitudinal and lateral control, enabling faster and more robust optimisation when tracking a reference line such as a crop row [53].

3.4 Learned Trajectory Proposals and Scoring

In the proposed hybrid architecture, a data-driven component may augment the classical planner for the local planning layer, responsible for short-horizon re-planning around unforeseen obstacles and selection among manoeuvre candidates. Imitation learning (IL) trains the local policy by mimicking expert driving demonstrations, providing a supervised-learning mechanism that avoids the sample-inefficiency and safety concerns of reinforcement learning [39, 40]. The learned component generates trajectory candidates or produces a cost map over the occupancy grid; the final trajectory selection is then performed by a safety filter that verifies collision freedom, kinematic feasibility, and geofence compliance before any candidate is passed to the controller [40, 41].

This architecture—where “ML proposes; the safety system validates”—appears consistently in the autonomous driving literature as the most practically sound approach for safety-critical systems [3, 40, 41]. Data-driven methods may also be used to learn a *cost map* from historical operational data, encoding terrain traversability, soil compaction risk, or energy consumption per unit area, which then influences the global planner’s swath sequencing [3].

3.5 Safety Validation

Regardless of whether the local planner is classical or learned, a dedicated safety validation layer is mandatory in agricultural machinery. This layer must enforce, as hard constraints: geofencing (vehicle remains within the declared operational boundary), emergency stop logic triggered by IMU shock, encoder stall, camera obstruction, or GNSS loss, and speed limiting based on operational context [3]. The safety supervisor must operate independently of the planning pipeline so that a failure in the learned component does not propagate to actuation [3].

The thesis system requires explicit strategies for each common failure mode: loss of RTK signal, camera blinding by dust or sun glare, and LiDAR fouling by mud. The documented recommendation is a hierarchical fallback: degrade from full-autonomy mode to reduced-speed odometry-only mode, then to safe-stop if confidence in

localisation falls below a threshold [3]. The Kalman filter and Extended Kalman Filter (EKF) frameworks are standard tools for maintaining state estimates during sensor outages by propagating IMU measurements forward [37, 38, 54].

3.6 Model Predictive Control for Trajectory Tracking

Model Predictive Control (MPC) is the standard trajectory tracking framework in high-precision autonomous vehicle applications due to its ability to enforce state and input constraints explicitly within the control loop [43, 47, 55]. The MPC solves, at each control cycle, a constrained finite-horizon optimisation:

$$\min_{u(0:H-1)} \sum_{k=0}^{H-1} \left(\|x_k - x_k^{\text{ref}}\|_Q^2 + \|u_k\|_R^2 + \|\Delta u_k\|_S^2 \right) \quad (1)$$

subject to the vehicle kinematic model $x_{k+1} = f(x_k, u_k)$, actuator limits $u_{\min} \leq u_k \leq u_{\max}$, and safety constraints $x_k \in \mathcal{X}_{\text{safe}}$ [2]. Here Q , R , and S are positive-definite weighting matrices trading off tracking accuracy, control effort, and rate of control change (smoothness). Nonlinear MPC (NMPC) variants, including decentralised and tube-based formulations, provide high-accuracy tracking for vehicles with nonlinear dynamics such as wheel slip or tractor-trailer articulation [47].

Hierarchical MPC architectures, in which a linearised MPC provides an initial solution that is refined by a nonlinear local adjustment layer, offer a practical balance between computational efficiency and constraint satisfaction accuracy [45, 46]. For the thesis embedded platform (Raspberry Pi 3B with a 100 ms control loop), a linearised kinematic bicycle MPC is recommended for the initial prototype, with NMPC reserved for a later hardware iteration with more computational resources [1, 3].

3.7 Agricultural Coverage Planning: Specific Considerations

Agricultural coverage planning imposes constraints that distinguish it from generic autonomous driving planning:

- **Implement geometry:** Each implement configuration (cutter, sprayer, plough) changes the effective working width, rear swing radius, and collision geometry. The planner must be parameterised by implement configuration [3].
- **Soil compaction:** Repeated traversal of the same soil strips degrades soil structure. Path planning should track and minimise overlap while achieving coverage targets [45, 46].
- **Crop protection:** Lateral deviation from swath centrelines must be bounded to avoid crop damage. Hybrid A* path-tracking variants incorporate this as a primary cost term [45, 46].
- **Energy and return-to-base:** Energy-aware planning with explicit battery models ensures that the vehicle completes its coverage task and returns to the charging station safely [56, 57].

- **Headland management:** Headland turns are frequently the largest source of non-working travel. Continuous-curvature headland planning is an active research topic [58].

The route optimisation objective for this thesis can be formalised, following the modelling approach proposed in the project design documents [3], as:

$$J = w_1 \cdot t + w_2 \cdot E + w_3 \cdot R + w_4 \cdot M + w_5 \cdot C \quad (2)$$

where t is total mission time, E is energy consumption, R is obstacle proximity risk, M is a penalty for complex or unnecessary manoeuvres, C is a soil compaction or crop-damage cost, and w_i are application-specific weights. The priority ordering for the thesis system is coverage quality, then energy efficiency, then reproducibility, safety, speed, and map-building [1]. Equation (2) must be supplemented with hard constraints on minimum turning radius, maximum speed, geofence, and minimum obstacle clearance, since soft weighting alone cannot guarantee safety [3].

3.8 Reinforcement Learning: Future Work, Not the Proposed Core

Reinforcement learning (RL) has been explored for agricultural coverage planning, including Soft Actor-Critic (SAC) formulations that balance coverage maximisation with energy constraints [56], and deep RL frameworks combining learned policies with probabilistic roadmaps for obstacle avoidance [2]. However, the thesis explicitly positions RL as a direction for future development rather than the core planning mechanism, for three reasons supported by the literature:

1. RL training requires large numbers of environment interactions; in agricultural settings these must occur primarily in simulation, introducing a sim-to-real gap that is difficult to close without extensive real-world validation data [39, 41].
2. Safety guarantees are difficult to establish for RL policies operating outside their training distribution, which is a hard requirement for machinery capable of causing physical harm [40, 41].
3. Classical coverage planners with well-understood convergence properties routinely achieve the target coverage rates ($\geq 98\%$) without the training overhead of RL [1, 51].

3.9 Evaluation Benchmarks for Planning

The nuPlan benchmark [59–61] provides a closed-loop planning evaluation framework based on 1,282 hours of real-world driving data across four cities and 73–75 distinct scenario types [59, 61]. Its Closed-Loop Score (CLS) aggregates safety (collision rate, off-road rate), comfort (acceleration and jerk), and efficiency (route progress, time-to-collision) into a single metric, with hard-multiplier penalties for collisions or

Table 2: Comparison of path planning approaches for agricultural coverage. Safety column refers to formal or empirical safety guarantees available. Ag. Suitability rates suitability for a thesis-level prototype.

Method	Type	Key Advantage	Safety	Thesis Suitability
Boustrophedon CPP [47]	Classical	Complete coverage guarantee, deterministic	High	Excellent baseline
Hybrid A* [45–47]	Graph search	Non-holonomic feasible; crop-row tracking	High	Good for headland turns
Lattice Planner [43]	Sampling	Structured, fast; kinematically consistent	High	Good for constrained spaces
RRT / RRT* [47]	Sampling	High-dimensional spaces; orchard navigation	Moderate	Useful in orchards
IL Trajectory Scoring [39, 40]	Learned	Adapts to expert patterns; no explicit model	Requires safety filter	Local planner enhancement
RL (SAC, DQN) [56]	Learned	End-to-end adaptability; energy awareness	Low without safety layer	Future work only
MPC Tracker [43, 47, 55]	Optimal Control	Constraint-aware; smooth; real-time	High	Essential for execution

illegal manoeuvres [60, 61]. Although nuPlan is road-vehicle specific, its evaluation philosophy—closed-loop interaction, scenario taxonomy, and composite metrics—is a useful template for designing agricultural planning evaluations. CARLA and the Bench2Drive benchmark [62] are noted for synthetic-environment evaluation but suffer from a simulation-to-real gap not present in nuPlan’s real-world logs [59, 62]. For the agricultural thesis, analogous closed-loop evaluation should be conducted in a physics-accurate agricultural simulator with synthetic field geometries and random obstacle placements.

4 Agricultural Computer Vision

4.1 Crop Recognition and Disease Identification

Deep learning has become the dominant methodology for automated plant disease identification, with Convolutional Neural Networks (CNNs) and, increasingly, Vision Transformers (ViTs) achieving high accuracy in image classification tasks [63–65]. The standard CNN architectures—ResNet, DenseNet, and EfficientNet—have been shown to identify disease patterns effectively in hierarchical feature extraction pipelines [66]. ViTs offer superior performance in capturing global spatial dependencies through self-attention mechanisms, which is advantageous for identifying small, irregular disease lesions scattered across a leaf surface [64, 67]. However, ViTs lack the inductive spatial bias inherent to CNNs and typically require larger annotated datasets to achieve robust generalisation [68, 69].

Hybrid CNN-ViT architectures exploit both capabilities: CNNs perform efficient local feature extraction for edges and textures, while ViT heads capture global contextual relationships [64, 70, 71]. Mixture of Experts (MoE) architectures integrated with ViT backbones use a gating mechanism to route inputs to specialised expert sub-networks, demonstrating improved adaptation to diverse environmental conditions such as varying lighting and disease severity [69]. Vision Mamba (ViM), replacing self-attention with selective state-space models for linear-time scaling, is an emerging architecture showing competitive accuracy-efficiency trade-offs for high-resolution UAV and edge-computing applications [70].

A persistent challenge is the *domain gap* between laboratory-controlled dataset conditions and real field conditions. Models achieve over 99% accuracy on PlantVillage under controlled settings but can fall to as low as 31% accuracy when evaluated on field-collected images [66, 72]. This dramatic degradation is largely attributable to *background bias*: models learn to respond to the uniform plain backgrounds of laboratory images rather than the actual biological disease symptoms, and fail when confronted with complex field backgrounds, occlusion, and illumination variation [66, 73].

4.2 Weed Classification, Detection, and Semantic Segmentation

Weed detection has evolved from image classification toward pixel-level semantic segmentation, which provides the spatial precision required for site-specific weed management (SSWM) and targeted herbicide application [74–76]. Encoder-decoder architectures based on U-Net remain foundational, with variants such as AgriResUpNet combining U-Net with residual learning to improve feature extraction and boundary demarcation in dense plant canopies [77]. Residual U-Net architectures have been shown to outperform Mask R-CNN-based models in dense plant environments [78].

Recent advances incorporate Transformer-based backbones (e.g., DPT with DINOv2) and Mamba-based architectures that capture long-range dependencies better than standard CNNs, with hybrid models combining Transformer modules and CNN backbones for improved structural consistency [75, 79]. For embedded deployment on autonomous agricultural robots (e.g., NVIDIA Jetson Xavier NX class hardware), lightweight architectures incorporating dual attention blocks (DA) and spatial connectivity attention (SCA) blocks balance accuracy with the inference speeds required for real-time operation [80]. Model pruning, quantisation via TensorRT, and efficient backbone selection are standard optimisation strategies for achieving the low latency required for real-time robotic weeding [75, 80].

A significant bottleneck in applied weed segmentation is the requirement for large, manually annotated datasets. Unsupervised labelling pipelines that exploit field row structure and robot pose information to automatically generate consistent training labels without human intervention represent a promising direction that directly addresses this bottleneck [81]. Uncertainty-aware models using evidential deep learning have been deployed to refine predictions for challenging or unknown vegetation components [81].

The integration of semantic segmentation with robotic actuation typically involves transforming detected weed coordinates from image space into a bird’s-eye view metric plane via homography, then scheduling a trajectory for the robotic spraying mechanism [82]. This perception-to-action pipeline is directly applicable to the thesis: BEV-registered crop-health detections feed the coverage planner’s task scheduling layer, enabling the vehicle to prioritise visit sequences for zones requiring intervention.

4.3 Key Datasets

4.3.1 PlantVillage

The PlantVillage dataset [83–85] consists of 54,306 high-quality RGB images of healthy and diseased plant leaves covering 14 crop species and up to 38 disease classes [83, 84, 86]. Introduced by Mohanty, Hughes, and Salathé, the dataset demonstrated that deep learning could achieve approximately 99.35% classification accuracy in identifying plant diseases under controlled settings [85, 86]. It has been widely used to benchmark transfer learning approaches, attention mechanisms, and hybrid architectures [85, 87]. The dataset is accessible through IEEE DataPort,

GitHub, and TensorFlow Datasets [83, 84, 88].

However, PlantVillage’s controlled uniform-background images introduce fundamental limitations for real-world deployment. Cross-environment generalisation tests have documented accuracy drops from laboratory to field conditions, with background bias identified as the primary mechanism [66, 72, 73]. Newer datasets including PlantWild and PlantSeg address these gaps by providing in-the-wild imagery and pixel-level segmentation masks [89, 90].

4.3.2 DeepWeeds

The DeepWeeds dataset [91, 92] comprises 17,509 images of eight weed species native to Australian rangelands, collected *in situ* across eight geographically distributed sites in northern Australia [91, 93]. Unlike PlantVillage, DeepWeeds captures plants in realistic environmental conditions with variable illumination, occlusion, rotation, scale, and background clutter [91, 93]. The original study established benchmarks using Inception-v3 (95.1% average accuracy) and ResNet-50 (95.7% average accuracy) [91, 93]. Deployment on an NVIDIA Jetson TX2 achieved 53.4 ms average inference time per image with ResNet-50, demonstrating feasibility for real-time robotic control [91, 93]. Subsequent semi-supervised methods such as Graph Weeds Net (GWN) have achieved 98.1% accuracy on the dataset [94]. DeepWeeds is primarily a classification dataset and does not provide pixel-level segmentation masks [95, 96]; segmentation-oriented tasks require supplemental datasets such as CropAndWeed, WE3DS (RGB-D), or RiceSEG [97, 98].

4.3.3 Agriculture-Vision

Agriculture-Vision [99, 100] is a large-scale aerial-imagery benchmark comprising 94,986 high-resolution images collected from 3,432 US farmlands between 2017 and 2019. Images are provided in both RGB and Near-Infrared (NIR) channels at ground sampling distances up to 10 cm/pixel, enabling computation of vegetation indices such as NDVI [99, 100]. Expert agronomists annotated nine categories of field anomaly patterns impacting crop yield, including weed clusters, water stress, nutrient deficiency, planter skips, and storm damage [99, 100]. The primary evaluation metric is mean Intersection-over-Union (mIoU), with the evaluation protocol adapted to handle multi-label pixels where anomaly patterns overlap [99, 100].

A 2023 extension of the dataset provides raw full-field imagery and 3,600 high-resolution images for self-supervised pre-training using architectures including Swin Transformer and CNN backbones. Label-free anomaly detection via Masked Image Modelling (SwinMAE with Anomaly Suppression Loss) has demonstrated a 6.3% mIoU improvement over prior supervised methods on this benchmark. Agriculture-Vision is particularly relevant to the thesis because its aerial perspective is conceptually aligned with the BEV representation produced by the perception layer, providing a basis for validating crop-health classification outputs registered in the BEV frame.

Table 3: Agricultural computer vision datasets relevant to the thesis. The thesis scenario involves ground-level (not aerial) imagery for the crop-health module.

Dataset	Size	Primary Task	Conditions	Key Limitation
PlantVillage [83, 84]	54,306 images; 14 crops; 38 classes	Leaf disease classification	Controlled lab; plain background	Severe lab- to-field gap; background bias [66, 72]
DeepWeeds [91, 92]	17,509 images; 8 weed species	Weed classifi- cation	In situ; re- alistic back- grounds	Classification only; no seg- mentation masks [95, 96]
Agriculture- Vision [99, 100]	94,986 aerial images; 3,432 fields	Anomaly segmentation (mIoU)	Aerial, multi- spectral (RGB+NIR)	Aerial per- spective; no ground-level view
CropAndWeed [97, 98]	Thousands an- notated	Crop/weed segmentation	Field condi- tions	Fewer species; limited ge- ographic diversity
WE3DS [97, 98]	RGB-D pairs	Semantic seg- mentation	Field with depth	Requires depth sensor

4.4 Transfer Learning and Domain Adaptation

Transfer learning from large-scale datasets (ImageNet pre-training) is the standard initialisation strategy for agricultural CV models, significantly reducing the quantity of domain-specific annotations required [85, 101]. However, transfer learning alone does not resolve the domain gap between controlled training conditions and field deployment: models must be explicitly adapted to the target environment [64, 66]. Domain adaptation techniques, including Cross-Domain Few-Shot Learning (CD-FSL), attempt to adapt models to new target domains with minimal labelled data [101]. Data augmentation strategies—including synthetic variation of lighting, background texture, occlusion, and image degradation—are widely used to artificially broaden the distribution seen during training [64, 66].

For the thesis, where ground-level crop images are available but field-scale BEV occupancy maps are not, a practical domain adaptation strategy involves: (1) pre-training the crop classification backbone on PlantVillage or DeepWeeds, (2) collecting a smaller set of field images from the operational environment, (3) fine-tuning the classification head on field images using augmentation to mitigate background bias, and (4) validating generalisation with held-out field images from different days, lighting conditions, and growth stages. This staged approach aligns with the broader recommendation in the agricultural CV literature for incremental data-centric improvement [66, 102].

4.5 Real-World Robustness Challenges

Field deployment of agricultural CV systems faces the following well-documented challenges, all of which must be addressed in the thesis experimental plan [3, 33, 34]:

- **Dust and sensor fouling:** Dust raised by the vehicle’s own passage can occlude cameras and LiDAR windows, particularly in dry conditions [3].
- **Illumination variability:** Shadow transitions, sun glare, and low-angle lighting create strong appearance shifts between frames [63, 64].
- **Seasonal appearance change:** Crop morphology changes substantially across growth stages, requiring models trained on one phenological stage to generalise to others [33, 34].
- **Occluded and overlapping plants:** Dense canopies create significant occlusion, requiring models capable of reasoning about partially visible plants [63, 64].
- **Edge cases:** Animals at rest, low-hanging branches, partially visible implements, and objects unseen during training represent the long tail of real-world variability [3].

The general recommendation is to assemble a dataset with variety across seasons, illumination conditions, and geographic locations, and to ensure that train and test splits are partitioned by field site or recording day rather than by random frame

sampling—the latter inflates measured accuracy by including near-duplicate frames in both splits [3].

5 Comparative Synthesis

5.1 Cross-Domain Coherence

The three thematic domains converge on a unified observation: *no single model or approach is sufficient*; robust performance requires modular, layered architectures that combine the generalisation power of learned representations with the verifiable safety of classical algorithms [2, 3, 42, 44]. This principle manifests differently in each domain:

In **BEV perception**, the combination of camera-based semantic features and LiDAR geometric features via attention-based fusion (BEVFusion [4, 12]) consistently outperforms either modality alone, and temporal fusion further improves dynamic object handling [11, 16]. In **hybrid planning**, the literature uniformly advocates classical coverage guarantors supplemented by learned local planners with verified safety filters [40, 45, 46]. In **agricultural CV**, transfer learning plus targeted domain adaptation outperforms either isolated laboratory training or naive field deployment [64, 101].

5.2 Tensions and Open Questions

A genuine tension exists between the computational demands of state-of-the-art BEV perception (which motivated the $40\times$ BEV pooling speedup in BEVFusion [4, 12]) and the resource constraints of agricultural embedded platforms (Raspberry Pi 3B, STM32) [1, 3]. Current literature does not provide a direct answer for running full multi-modal BEV transformers on such hardware; the realistic path is (a) using lightweight variants such as TPVFormer [25, 30] or SparseOcc [31], (b) offloading inference to an edge GPU (NVIDIA Jetson class), or (c) accepting a simplified occupancy representation based on lightweight depth estimation. The thesis should explicitly characterise this trade-off rather than making unsupported performance claims.

A second tension concerns dataset coverage: the agricultural CV datasets most relevant to the thesis task (PlantVillage, DeepWeeds) were collected under conditions that differ substantially from the operational scenario [66, 91], while Agriculture-Vision provides agronomically relevant labels but from an aerial perspective not directly aligned with the ground-level camera [99, 100]. Bridging this gap requires the simulation-plus-real-data strategy described in the experimental methodology section.

6 Proposed Modular Research Architecture

6.1 Architecture Overview

Based on the synthesis of the three thematic domains and the project design specifications [1–3], the proposed thesis architecture comprises five functional layers organised as shown in Figure 2. This architecture is appropriate for an academic prototype and does not claim production-grade performance.

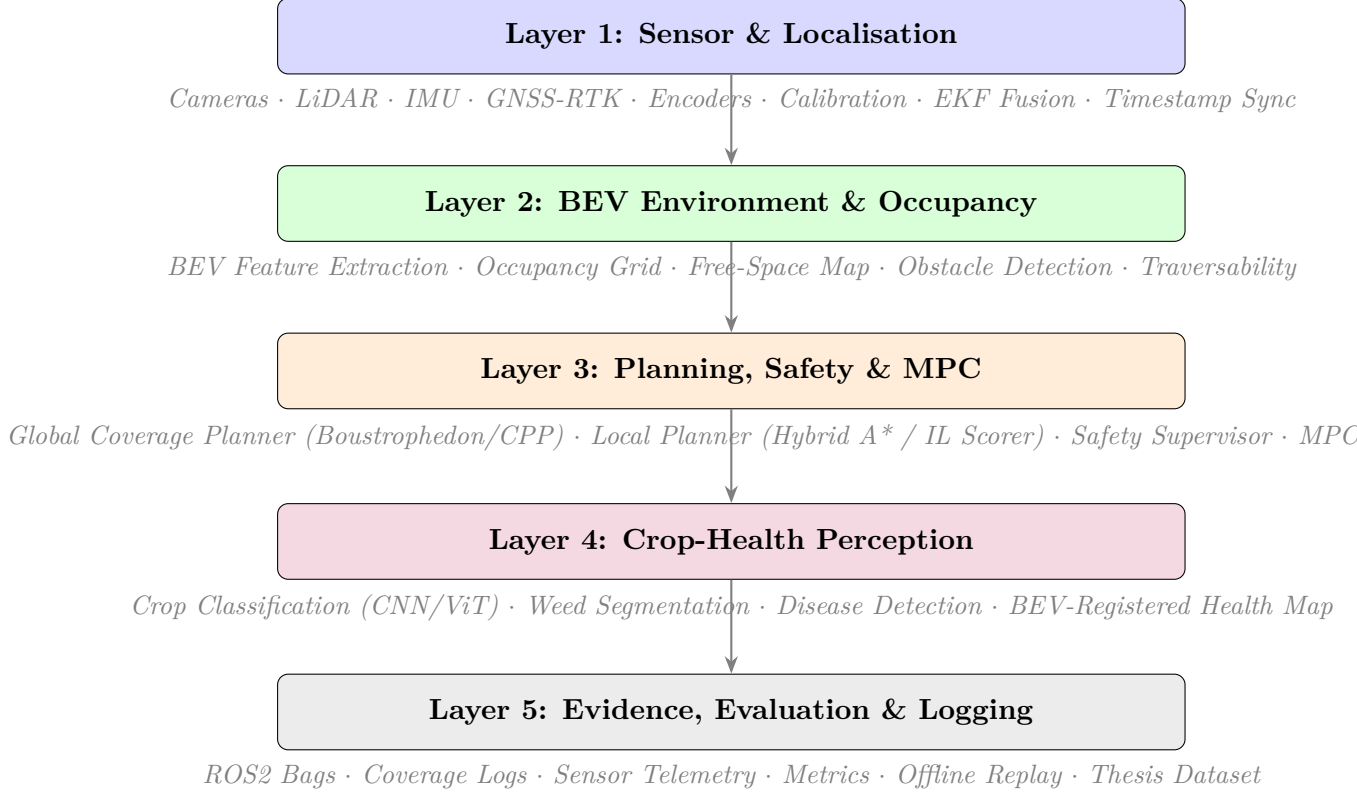


Figure 2: Proposed five-layer modular architecture for the thesis prototype. Arrows denote primary data flows; each layer is independently testable and corresponds to a vertical development slice.

6.2 Layer 1: Sensor and Localisation

This layer is responsible for acquiring, time-stamping, and fusing all raw sensor streams into a coherent ego-state estimate. The sensor suite comprises dual cameras (frontal and rear, 30 FPS HD), a frontal LiDAR, an IMU operating at ≥ 100 Hz, wheel encoders, and GNSS-RTK [1]. An EKF fuses IMU measurements, wheel odometry, and periodic GNSS corrections to maintain a continuous pose estimate robust to brief GNSS outages [37, 38, 54]. Accurate extrinsic calibration (camera–LiDAR–IMU) is a prerequisite for all downstream processing; the literature strongly identifies calibration quality as the primary practical barrier to system performance [3, 33]. ROS2 is the recommended middleware, providing standardised topic interfaces, timestamped bag recording, and replay capability [1].

6.3 Layer 2: BEV Environment and Occupancy

Layer 2 converts the fused sensor streams into a metric BEV occupancy representation. For the initial prototype, a simplified occupancy grid derived from LiDAR point-cloud projection and camera-based depth estimation (using a lightweight encoder such as TPVFormer [25, 30] or a depth-estimation network) is realistic given the computational constraints of the Raspberry Pi 3B [1]. Full multi-modal transformer-based BEV perception (BEVFusion [4, 12]) is the aspirational target for a subsequent hardware upgrade to a NVIDIA Jetson-class edge computer. The occupancy grid resolution should match the vehicle footprint (0.1 m cells for a 1×1.8 m vehicle) to enable precise collision checking [1]. The layer exports: (a) a binary traversability map, (b) a semantic obstacle map with class labels (tree, wall, animal, unknown), and (c) confidence estimates for each cell.

6.4 Layer 3: Planning, Safety, and MPC

This is the operational core of the thesis system. The global planner implements Boustrophedon CPP parameterised by implement working width and target overlap fraction [1, 47]. Frontier-based fill-in resolves uncovered cells after the main swath pass [1]. A Hybrid A* or lattice-based local planner handles real-time obstacle avoidance and short-horizon re-planning [45–47]. Optionally, an imitation-learned trajectory scorer augments the local planner by ranking candidate trajectories according to patterns learned from expert demonstrations [39, 40]. The safety supervisor layer—operating independently of the planning pipeline—enforces geofencing, emergency stop, speed limiting, and sensor-failure fallbacks [3]. The MPC tracker (linearised kinematic bicycle model) executes the approved trajectory at 10 Hz, respecting actuator limits and curvature constraints [2, 55].

6.5 Layer 4: Crop-Health Perception

Layer 4 implements the crop-health identification capability that distinguishes the thesis from a generic coverage robot. A lightweight CNN or hybrid CNN-ViT architecture [64, 70] performs per-frame classification of crop condition (healthy, diseased, weed-infested) using images from the frontal camera. Classification outputs are registered to the BEV frame using vehicle pose from Layer 1 and projected depth, producing a geo-referenced health map that persists across the coverage pass. This health map feeds back into Layer 3’s route cost function (Equation 2): zones with detected anomalies receive lower traversal cost, biasing the planner to revisit them for confirmation or targeted intervention [3]. The initial model is trained on PlantVillage and DeepWeeds, with domain adaptation using field images from the operational environment [66, 101].

6.6 Layer 5: Evidence, Evaluation, and Logging

All sensor streams, planning decisions, coverage states, and crop-health detections are logged as ROS2 bags and structured CSV files [1]. This logging layer serves a

dual purpose: it enables offline replay for debugging and hyperparameter tuning, and it constitutes the thesis dataset for reporting reproducible results. Metrics logged include: cell-level coverage fraction and overlap fraction, distance travelled, energy consumed, obstacle detection events, safety supervisor intervention count, and per-frame crop-health classification confidence. The incremental documentation of results within each development slice avoids a compressed documentation sprint at the end of the project [1].

7 Experimental Methodology and Evaluation Design

7.1 The Challenge: Crop Images Available, Field BEV Maps Absent

The most significant experimental constraint of the thesis is the absence of field-scale BEV occupancy ground-truth data. Crop-disease images are available from PlantVillage [83, 84] and DeepWeeds [91, 92], but neither dataset provides the spatially-registered BEV representations needed to validate the perception and planning pipeline. This asymmetry necessitates a staged experimental design with different validation strategies for each module.

7.2 Stage 1: Simulation-First Validation

The recommended first stage is simulation using Gazebo with ROS2 integration, providing a physics-accurate environment for the vehicle model and sensor simulators [1]. The simulator enables:

- Automated generation of synthetic annotated BEV maps with known ground truth for obstacle positions and free-space extent [1].
- Systematic variation of obstacle layouts (number, positions) for coverage planner stress-testing across 10–20 randomly generated scenarios.
- Domain randomisation of lighting, camera noise, and sensor drift to broaden the training distribution for perception models [3].
- Replay of recorded trajectories for offline analysis and metric computation without physical vehicle operation.

Acceptance criteria at the simulation stage follow the project plan [1]: $\geq 95\%$ overlap before hardware integration, rising to $\geq 98\%$ overlap after system tuning.

7.3 Stage 2: Domain Validation for Crop-Health Module

For the crop-health perception module, a separate experimental protocol is required:

1. **Pre-training evaluation:** Train CNN/ViT backbone on PlantVillage [83, 84] and report accuracy on the PlantVillage test set as the baseline. Report separately on a held-out set of field images to document the lab-to-field gap [66, 72].
2. **Fine-tuning evaluation:** Collect a field image set from the operational environment (minimum recommended: 200–500 annotated images per class). Fine-tune the classification head and measure accuracy improvement. Partition train/test splits by recording day, not by random frame sampling, to prevent data leakage [3].
3. **Inference latency evaluation:** Report per-frame classification latency on the Raspberry Pi 3B target hardware. The target is ≤ 50 ms per frame to fit within the 100 ms control loop [1].
4. **Domain stress-test:** Evaluate model performance across different times of day, with simulated dust occlusion, and on leaves at different growth stages to characterise robustness boundaries [63, 64].

7.4 Stage 3: Real-Hardware Integration Testing

Integration testing on the physical platform follows the vertical-slice approach described in the project plan [1]:

1. **Sensor fusion validation:** Measure odometry drift over a 30-minute real-world run (target: $< 5\%$ of path length). Validate GNSS-IMU-encoder fusion using ground-truth positions from RTK-GNSS [1].
2. **Coverage planner validation:** Execute coverage passes on a simplified test lawn and measure achieved coverage using manual grid overlay or structured visual inspection. Target: $\geq 95\%$ coverage at first integration, rising to $\geq 98\%$ after tuning [1].
3. **Obstacle avoidance validation:** Introduce known obstacles (cones or cylinders at calibrated positions) and verify zero collision events across repeated runs.
4. **Safety supervisor validation:** Intentionally induce each failure mode (simulated GNSS loss, camera occlusion, encoder stall) and verify correct fallback behaviour within a 1-second response time [3].

7.5 Recommended Metrics

7.6 Synthetic Data and Simulation Strategy

Given the absence of field BEV ground truth, synthetic data generation in the Gazebo simulator serves as the primary source of training data for the occupancy and traversability modules. The recommended procedure is:

Table 4: Recommended evaluation metrics for the thesis, organised by system layer. All metrics should be reported with mean and standard deviation across multiple runs.

Layer	Metric	Definition and Target
Planning	Coverage fraction	Fraction of operational area traversed; target ≥ 0.98
Planning	Overlap fraction	Fraction of area traversed more than once; target ≤ 0.02
Planning	Path efficiency	Total distance / theoretical minimum Boustrophedon distance
Localisation	Odometry drift	Absolute pose error at end of run / total path length; target $< 5\%$
BEV Perception	Occupancy mIoU	Mean IoU over free/occupied/unknown classes (simulation); following [31]
Crop-Health	Classification accuracy	Per-class accuracy on held-out field test set
Crop-Health	Inference latency	Per-frame latency on target hardware (ms); target ≤ 50 ms [1]
Safety	Intervention rate	Number of safety supervisor activations per km of operation [38, 103]
Safety	False stop rate	Spurious emergency stops not triggered by actual hazard
System	Battery autonomy	Operating time at field speed; target ≥ 2 h [1]

- Generate 10,000–20,000 synthetic image frames with automatically labelled occupancy masks by varying obstacle layouts, lighting conditions, and camera orientations within the simulator [1].
- Apply domain randomisation (surface texture, lighting angle, camera noise model, simulated dust) to reduce the sim-to-real gap [3].
- Use the synthetic dataset to train the occupancy estimation component, then fine-tune on a smaller set of real-world observations collected during Stage 3 integration testing.
- Validate the domain transfer by measuring occupancy mIoU on real-world frames for which ground truth is available from LiDAR point clouds [31].

This simulation-first, real-world-fine-tuning strategy is consistent with the agricultural robotics literature’s recommendation to accelerate development with synthetic data while closing product quality with a smaller real dataset [3].

7.7 Ablation and Baseline Comparisons

Academically rigorous evaluation requires comparison against well-defined baselines and ablation of individual components:

- **Coverage planning baseline:** Random walk coverage and spiral pattern coverage should be compared against the Boustrophedon planner on coverage fraction and path efficiency metrics.
- **Perception ablation:** Evaluate the planning system with (a) no obstacle perception (GNSS-only), (b) geometric LiDAR-only obstacle detection, and (c) the full camera-LiDAR BEV occupancy system, to quantify each perception component’s contribution.
- **Crop-health module ablation:** Compare coverage planning with and without crop-health-guided route weighting to measure the operational benefit of the health perception layer.
- **Temporal fusion ablation:** Compare single-frame versus temporal BEV representations for occupancy estimation to characterise the value of temporal context in the agricultural setting.

These ablations allow the thesis to make quantitatively supported claims about which components contribute most to system performance—a requirement for academic evaluation [1].

8 Limitations and Threats to Validity

8.1 Domain Gap from Road-Vehicle BEV Literature

The primary threat to the validity of applying BEV perception architectures to the agricultural domain is the large domain gap between road-vehicle benchmarks (nuScenes [27, 29], Occ3D [28]) and the agricultural operational environment. Agricultural scenes feature repetitive texture (crop rows), seasonally variable appearance, and object categories absent from road datasets [33, 34]. Performance numbers from nuScenes evaluations of BEVFormer [5, 6] or BEVFusion [4, 12] are not transferable to the agricultural setting without domain adaptation. The thesis should present agricultural BEV performance as separate from—and not compared to—road-vehicle benchmarks.

8.2 Simulator Fidelity and Sim-to-Real Gap

The simulator provides a controlled environment for initial development but introduces a sim-to-real gap in sensor noise models, surface dynamics, and visual appearance. Odometry drift measured in simulation may differ substantially from real-world behaviour due to wheel slip on soft soil, IMU vibration from the cutting mechanism, and dust interference with camera images [1, 3]. Early real-world testing (from Stage 3 onward) is essential to characterise this gap; the vertical-slice development plan [1] schedules real-world field tests within the first hardware integration phase.

8.3 Dataset Limitations

The crop-health perception module relies on PlantVillage [83, 84] and DeepWeeds [91, 92] for initial training, both of which have well-documented limitations for real-field deployment [66, 72, 93]. PlantVillage’s controlled backgrounds induce background bias [66, 73]; DeepWeeds covers only eight species specific to Australian rangelands [91, 93]. The thesis should explicitly acknowledge that the crop-health module’s performance is bounded by the availability of field-annotated data from the specific operational environment, and that reported accuracy figures on PlantVillage test sets are upper bounds for field performance.

8.4 Computational Resource Constraints

The target hardware (Raspberry Pi 3B) provides 1 GB RAM and a quad-core Cortex-A53 CPU without hardware neural-network acceleration [1]. Full BEVFusion inference [4, 12] or unoptimised transformer-based crop classification is infeasible in real time on this platform. The prototype must use lightweight or quantised models; all reported inference latencies must be measured on the actual target hardware rather than inferred from desktop or GPU benchmarks [1, 18]. Quantisation from floating-point to INT8 precision introduces known accuracy trade-offs, and the

magnitude of this degradation for the specific models and tasks must be characterised experimentally.

8.5 Single-Vehicle Prototype Scope

The thesis prototype addresses single-vehicle autonomous operation on a known field geometry with static obstacles. Dynamic obstacles (animals, farm workers), multi-vehicle coordination, and automatic implement docking are out of scope [2]. Generalisation claims must be bounded accordingly; the thesis should not extrapolate prototype results to fleets, unknown terrains, or fully dynamic environments.

9 Research Gaps and Thesis Contributions

9.1 Identified Research Gaps

The synthesis of the literature reveals the following gaps directly relevant to the proposed thesis:

1. **Agricultural BEV benchmarks:** There is no publicly available BEV occupancy dataset for agricultural ground vehicles analogous to Occ3D-nuScenes [28]. The thesis will contribute a simulation-generated dataset as a partial remedy.
2. **Lightweight BEV perception for embedded agricultural platforms:** The embedded-deployment literature for PETR-family models [18, 19] addresses automotive-grade embedded processors; equivalent analysis for Raspberry Pi or STM32 class hardware in agricultural settings is absent.
3. **Crop-health BEV integration:** Existing crop-health perception work operates on individual images or aerial frames [99, 100]; integration with a BEV map maintained by a ground vehicle in real time is not addressed in the reviewed literature.
4. **Hybrid CPP with learned local planning in agriculture:** The imitation-learning-augmented hybrid planning architecture reviewed in the planning literature [39, 40] has not been applied to agricultural coverage with simultaneous crop-health monitoring.
5. **Domain adaptation protocols for agricultural CV:** Systematic protocols for adapting road-vehicle BEV models to agricultural scenes—including benchmark evaluation procedures—are not established.

9.2 Research Questions

The thesis addresses the following research questions:

1. **RQ1:** Can a lightweight BEV occupancy representation, derived from camera and LiDAR data on an embedded platform, provide sufficient scene understanding for safe agricultural coverage planning?
2. **RQ2:** Does a hybrid classical-plus-learned planning architecture (Boustrophedon CPP with Hybrid A* local planning and MPC tracking) achieve the target coverage ($\geq 98\%$) specifications in real-world agricultural field conditions?
3. **RQ3:** Can a crop-health classifier trained on PlantVillage and DeepWeeds, domain-adapted to field conditions, be deployed in real time on the target hardware (≤ 50 ms inference latency) and produce useful health maps registered in the vehicle BEV frame?
4. **RQ4:** How does health-map-guided route weighting in the coverage planner affect operational efficiency compared to pure geometric coverage without perception-guided prioritisation?

9.3 Thesis Contributions

The anticipated contributions of the thesis, stated proportionately to the academic prototype scope, are:

1. A **validated modular architecture** for hybrid BEV-based autonomous agricultural navigation, integrating five distinct functional layers and demonstrated in simulation and on physical hardware.
2. A **lightweight BEV occupancy pipeline** characterised for embedded deployment, including quantitative measurement of the sim-to-real domain gap in the agricultural setting.
3. A **crop-health BEV integration method** that registers ground-level crop-health classifications into a vehicle-maintained BEV map and uses this map to influence coverage planning.
4. A **reproducible experimental dataset and evaluation protocol** comprising simulation logs, real-world sensor bags, crop classification results, and coverage metric records, enabling future researchers to build on and compare against the thesis results.
5. A **documented staged research plan** demonstrating how the vertical-slice development methodology [1] integrates hardware, firmware, software, and machine learning development in a constrained 6-month timeline.

10 Conclusion

This literature review has examined the foundational and current-state-of-the-art research in three domains essential to the proposed thesis: multimodal BEV transformer perception, hybrid path planning for autonomous agricultural vehicles, and agricultural computer vision. The review identifies a clear convergence across domains:

robust autonomous systems require modular layered architectures that combine the generalisation capacity of learned representations with the verifiable guarantees of classical algorithms and control-theoretic methods.

BEVFormer [5, 6], BEVFusion [4, 12], BEVDet4D [9, 10], and PETR [13, 14] establish the state of the art in BEV perception on road-vehicle benchmarks, but their direct transfer to agricultural settings is subject to significant domain gaps requiring systematic adaptation. Classical coverage planners with Boustrophedon decomposition [47], augmented by Hybrid A* [45, 46] for local replanning and MPC [47, 55] for trajectory tracking, provide the reliable operational backbone for the thesis system, with imitation-learned local trajectory scoring as an optional enhancement [39, 40]. Reinforcement learning is identified as a future research direction, not a component of the proposed prototype [1–3]. Crop-health perception using CNN and hybrid CNN-ViT architectures trained on PlantVillage [83, 84] and DeepWeeds [91, 92], with domain adaptation for the operational field environment, provides the agricultural differentiation of the system compared to a generic coverage robot.

The proposed five-layer modular architecture, staged experimental methodology, and dataset strategy are calibrated to the constraints of an academic prototype: crop images available but field BEV maps absent, an embedded computational platform, and a six-month development timeline [1]. The review explicitly distinguishes established research results from claims that must be validated experimentally, and identifies four research questions and five thesis contributions proportionate to the prototype scope. This foundation positions the thesis to make a credible and reproducible contribution to the intersection of autonomous vehicle perception, agricultural robotics, and precision crop monitoring.

References

- [1] *ProjectPlanning.pdf*. Uploaded document.
- [2] *MVP_TFM.pdf*. Uploaded document.
- [3] *ML_Models.pdf*. Uploaded document.
- [4] *arxiv.org*. URL: <https://arxiv.org/html/2205.13542v3> (visited on 08/19/2026).
- [5] *arxiv.org*. URL: <https://arxiv.org/abs/2203.17270> (visited on 08/19/2026).
- [6] *ar5iv.labs.arxiv.org*. URL: <https://ar5iv.labs.arxiv.org/html/2203.17270> (visited on 08/19/2026).
- [7] *arxiv.org*. URL: <https://arxiv.org/html/2604.02930v1> (visited on 08/19/2026).
- [8] *arxiv.org*. URL: <https://arxiv.org/html/2405.04299v2> (visited on 08/19/2026).
- [9] *arxiv.org*. URL: <https://www.arxiv.org/pdf/2405.16099> (visited on 08/19/2026).

- [10] *arxiv.org*. URL: <https://arxiv.org/html/2402.18140v1> (visited on 08/19/2026).
- [11] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12063725/> (visited on 08/19/2026).
- [12] *ar5iv.labs.arxiv.org*. URL: <https://ar5iv.labs.arxiv.org/html/2205.13542> (visited on 08/19/2026).
- [13] *arxiv.org*. URL: <https://arxiv.org/abs/2203.05625> (visited on 08/19/2026).
- [14] *arxiv.org*. URL: <https://arxiv.org/pdf/2203.05625> (visited on 08/19/2026).
- [15] *arxiv.org*. URL: <https://arxiv.org/abs/2402.18140> (visited on 08/19/2026).
- [16] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12470275/> (visited on 08/19/2026).
- [17] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10611098/> (visited on 08/19/2026).
- [18] *arxiv.org*. URL: <https://arxiv.org/html/2511.09347v1> (visited on 08/19/2026).
- [19] *arxiv.org*. URL: <https://arxiv.org/html/2502.15488v1> (visited on 08/19/2026).
- [20] *arxiv.org*. URL: <https://arxiv.org/pdf/2205.13542> (visited on 08/19/2026).
- [21] *arxiv.org*. URL: <https://arxiv.org/pdf/2205.13542v1> (visited on 08/19/2026).
- [22] *arxiv.org*. URL: <https://arxiv.org/abs/2205.13542> (visited on 08/19/2026).
- [23] *arxiv.org*. URL: <https://arxiv.org/html/2605.11799> (visited on 08/19/2026).
- [24] *arxiv.org*. URL: <https://arxiv.org/html/2504.12643> (visited on 08/19/2026).
- [25] *arxiv.org*. URL: <https://arxiv.org/html/2405.02595v1> (visited on 08/19/2026).
- [26] *arxiv.org*. URL: <https://arxiv.org/html/2407.13155v2> (visited on 08/19/2026).
- [27] *nuscenes.org*. URL: <https://www.nuscenes.org/> (visited on 08/19/2026).
- [28] *arxiv.org*. URL: <https://arxiv.org/html/2512.02448v1> (visited on 08/19/2026).
- [29] *nuscenes.org*. URL: <https://www.nuscenes.org/object-detection> (visited on 08/19/2026).
- [30] *arxiv.org*. URL: <https://arxiv.org/html/2412.07163v1> (visited on 08/19/2026).
- [31] *arxiv.org*. URL: <https://arxiv.org/html/2312.17118v4> (visited on 08/19/2026).
- [32] *arxiv.org*. URL: <https://arxiv.org/html/2411.12177v1> (visited on 08/19/2026).
- [33] *mdpi.com*. URL: <https://www.mdpi.com/2077-0472/15/12/1297> (visited on 08/19/2026).

- [34] *roboticsproceedings.org*. URL: <https://www.roboticsproceedings.org/rss17/p019.pdf> (visited on 08/19/2026).
- [35] *frontiersin.org*. URL: <https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2025.1597912/full> (visited on 08/19/2026).
- [36] *doi.org*. URL: <https://doi.org/10.3390/agronomy16141310> (visited on 08/19/2026).
- [37] *arxiv.org*. URL: <https://arxiv.org/html/2411.10974v1> (visited on 08/19/2026).
- [38] *ar5iv.labs.arxiv.org*. URL: <https://ar5iv.labs.arxiv.org/html/2106.15029> (visited on 08/19/2026).
- [39] *arxiv.org*. URL: <https://arxiv.org/html/2503.23650v1> (visited on 08/19/2026).
- [40] *arxiv.org*. URL: <https://arxiv.org/abs/2512.11944> (visited on 08/19/2026).
- [41] *intechopen.com*. URL: <https://www.intechopen.com/chapters/1206990> (visited on 08/19/2026).
- [42] *frontiersin.org*. URL: <https://www.frontiersin.org/journals/neurorobotics/articles/10.3389/fnbot.2025.1451923/full> (visited on 08/19/2026).
- [43] *mdpi.com*. URL: <https://www.mdpi.com/1424-8220/24/15/4808> (visited on 08/19/2026).
- [44] *arxiv.org*. URL: <https://arxiv.org/html/2406.05575v1> (visited on 08/19/2026).
- [45] *arxiv.org*. URL: <https://arxiv.org/html/2411.14086v1> (visited on 08/19/2026).
- [46] *arxiv.org*. URL: <https://arxiv.org/html/2411.14086v2> (visited on 08/19/2026).
- [47] *annualreviews.org*. URL: <https://www.annualreviews.org/doi/10.1146/annurev-control-053018-023617> (visited on 08/19/2026).
- [48] *arxiv.org*. URL: <https://arxiv.org/html/2412.19813v1> (visited on 08/19/2026).
- [49] *researchgate.net*. URL: https://www.researchgate.net/publication/390360515_A_Comprehensive_Survey_on_Coverage_Path_Planning_for_Mobile_Robots_in_Dynamic_Environments (visited on 08/19/2026).
- [50] *doi.org*. URL: <https://doi.org/10.3390/agronomy15122793> (visited on 08/19/2026).
- [51] *mdpi.com*. URL: <https://www.mdpi.com/2077-0472/16/1/64> (visited on 08/19/2026).
- [52] *link.springer.com*. URL: <https://link.springer.com/article/10.1007/s42154-023-00222-7> (visited on 08/19/2026).
- [53] *frontiersin.org*. URL: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2024.1403385/full> (visited on 08/19/2026).

- [54] *arxiv.org*. URL: <https://arxiv.org/html/2403.17774v3> (visited on 08/19/2026).
- [55] *sciencedirect.com*. URL: <https://www.sciencedirect.com/science/article/pii/S2215098625000059> (visited on 08/19/2026).
- [56] *arxiv.org*. URL: <https://arxiv.org/html/2601.16405> (visited on 08/19/2026).
- [57] *doi.org*. URL: <https://doi.org/10.3390/agriculture15080882> (visited on 08/19/2026).
- [58] *onlinelibrary.wiley.com*. URL: <https://onlinelibrary.wiley.com/doi/full/10.1002/rob.22489> (visited on 08/19/2026).
- [59] *arxiv.org*. URL: <https://arxiv.org/html/2403.04133v1> (visited on 08/19/2026).
- [60] *emergentmind.com*. URL: <https://www.emergentmind.com/topics/nuplan-benchmark-52dc041d-e596-4342-b1b6-f4ffc15b905a> (visited on 08/19/2026).
- [61] *emergentmind.com*. URL: <https://www.emergentmind.com/topics/nuplan> (visited on 08/19/2026).
- [62] *openreview.net*. URL: <https://openreview.net/pdf?id=CdGvA5qE67> (visited on 08/19/2026).
- [63] *frontiersin.org*. URL: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2025.1538163/full> (visited on 08/19/2026).
- [64] *mdpi.com*. URL: <https://www.mdpi.com/2313-433X/11/10/326> (visited on 08/19/2026).
- [65] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10894121/> (visited on 08/19/2026).
- [66] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC13066816/> (visited on 08/19/2026).
- [67] *sciencedirect.com*. URL: <https://www.sciencedirect.com/science/article/pii/S2772375525007221> (visited on 08/19/2026).
- [68] *arxiv.org*. URL: <https://arxiv.org/html/2504.21706v4> (visited on 08/19/2026).
- [69] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12213485/> (visited on 08/19/2026).
- [70] *sciencedirect.com*. URL: <https://www.sciencedirect.com/science/article/abs/pii/S0168169926001468> (visited on 08/19/2026).
- [71] *dl.acm.org*. URL: <https://dl.acm.org/doi/10.1145/3659677.3659685> (visited on 08/19/2026).
- [72] *nature.com*. URL: <https://www.nature.com/articles/s41598-026-45464-7> (visited on 08/19/2026).
- [73] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11466843/> (visited on 08/19/2026).
- [74] *frontiersin.org*. URL: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2025.1746406/full> (visited on 08/19/2026).

- [75] *sciencedirect.com*. URL: <https://www.sciencedirect.com/science/article/pii/S2772375526001188> (visited on 08/19/2026).
- [76] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12827514/> (visited on 08/19/2026).
- [77] *iieta.org*. URL: <https://iieta.org/journals/isi/paper/10.18280/isi.290516> (visited on 08/19/2026).
- [78] *agroengineering.org*. URL: <https://www.agroengineering.org/jae/article/view/1741> (visited on 08/19/2026).
- [79] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12609258/> (visited on 08/19/2026).
- [80] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10768065/> (visited on 08/19/2026).
- [81] *frontiersin.org*. URL: <https://www.frontiersin.org/journals/robotics-and-ai/articles/10.3389/frobt.2025.1548143/full> (visited on 08/19/2026).
- [82] *mdpi.com*. URL: <https://www.mdpi.com/2077-0472/16/7/810> (visited on 08/19/2026).
- [83] *ieee-dataport.org*. URL: <https://ieee-dataport.org/documents/plantvillage-plant-disease-classification-dataset> (visited on 08/19/2026).
- [84] *github.com*. URL: <https://github.com/spmohanty/plantvillage-dataset> (visited on 08/19/2026).
- [85] *ijfmr.com*. URL: <https://www.ijfmr.com/papers/2026/1/68968.pdf> (visited on 08/19/2026).
- [86] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC5032846/> (visited on 08/19/2026).
- [87] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9798446/> (visited on 08/19/2026).
- [88] *tensorflow.org*. URL: https://www.tensorflow.org/datasets/catalog/plant_village (visited on 08/19/2026).
- [89] *arxiv.org*. URL: <https://arxiv.org/html/2408.03120v1> (visited on 08/19/2026).
- [90] *nature.com*. URL: <https://www.nature.com/articles/s41597-025-06513-4> (visited on 08/19/2026).
- [91] *nature.com*. URL: <https://www.nature.com/articles/s41598-018-38343-3> (visited on 08/19/2026).
- [92] *github.com*. URL: <https://github.com/AlexOlsen/DeepWeeds> (visited on 08/19/2026).
- [93] *researchonline.jcu.edu.au*. URL: <https://researchonline.jcu.edu.au/60792/1/s41598-018-38343-3.pdf> (visited on 08/19/2026).
- [94] *pmc.ncbi.nlm.nih.gov*. URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8197187/> (visited on 08/19/2026).

- [95] *researchgate.net*. URL: https://www.researchgate.net/publication/331096007_DeepWeeds_A_Multiclass_Weed_Species_Image_Dataset_for_Deep_Learning (visited on 08/19/2026).
- [96] *arxiv.org*. URL: <https://arxiv.org/html/2406.00313v1> (visited on 08/19/2026).
- [97] *github.com*. URL: <https://github.com/vicdxxx/Weed-Datasets-Survey-2023> (visited on 08/19/2026).
- [98] *github.com*. URL: <https://github.com/ricber/digital-agriculture-datasets> (visited on 08/19/2026).
- [99] *agriculture-vision.com*. URL: <https://www.agriculture-vision.com/agriculture-vision-2020/dataset> (visited on 08/19/2026).
- [100] *openaccess.thecvf.com*. URL: https://openaccess.thecvf.com/content_CVPR_2020/html/Chiu_Agriculture-Vision_A_Large_Aerial_Image_Database_for_Agricultural_Pattern_Analysis_CVPR_2020_paper.html (visited on 08/19/2026).
- [101] *frontiersin.org*. URL: <https://www.frontiersin.org/journals/plant-science/articles/10.3389/fpls.2024.1434222/xml/nlm> (visited on 08/19/2026).
- [102] *sciencedirect.com*. URL: <https://www.sciencedirect.com/science/article/pii/S2773186326001003> (visited on 08/19/2026).
- [103] *arxiv.org*. URL: <https://arxiv.org/html/2404.17718> (visited on 08/19/2026).